

VIDYAPEETH



BATCH CODE: 41-AJ222^{MA} 202⁶-2⁷

SUBJECT NAME: Mathematics

**CHAPTER NAME:
Sequence and Series**



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Lecture No.

12



Today's Goal



Subtopic

Special Series

Subtopic

Telescopic Series



Topic: Quick Last Lecture Revision –



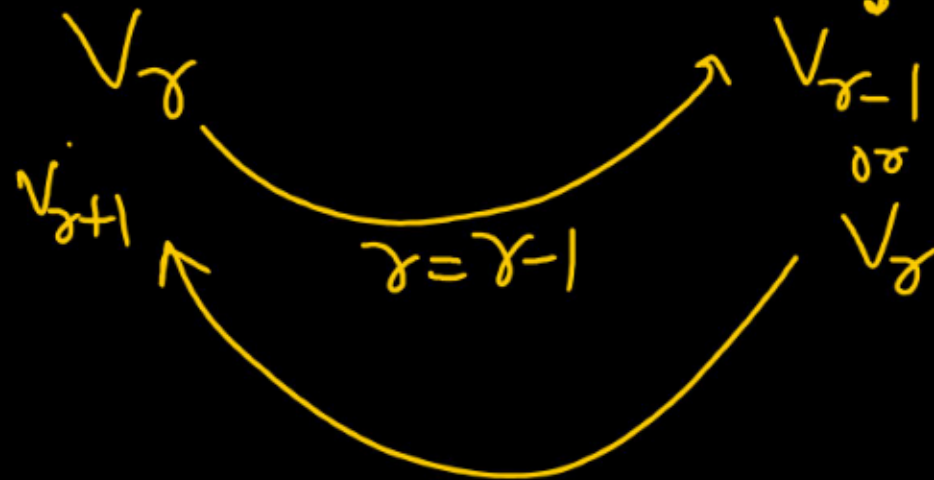
$$S = 1 \cdot 2 \cdot 3 \cdot 4 + 2 \cdot 3 \cdot 4 \cdot 5 + 3 \cdot 4 \cdot 5 \cdot 6 + \dots \text{ upto } n \text{ terms}$$

$$T_r = r \cdot (r+1) \cdot (r+2) \cdot (r+3)$$

$$T_r = \frac{1}{5} \cdot r \cdot (r+1) \cdot (r+2) \cdot (r+3) \cdot 5$$

$$T_r = \frac{1}{5} \cdot r \cdot (r+1) \cdot (r+2) \cdot (r+3) \cdot [(r+4) - (r-1)]$$

$$T_r = \frac{1}{5} \cdot [r \cdot (r+1) \cdot (r+2) \cdot (r+3) \cdot (r+4) - (r-1) \cdot r \cdot (r+1) \cdot (r+2) \cdot (r+3)]$$



$$T_1 = \frac{1}{5} [1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 - \cancel{0 \cdot 1 \cdot 2 \cdot 3 \cdot 4}]$$

$$T_2 = \frac{1}{5} [2 \cdot 3 \cdot 4 \cdot 5 \cdot 6 - \cancel{1 \cdot 2 \cdot 3 \cdot 4 \cdot 5}]$$

$$T_3 = \frac{1}{5} [3 \cdot 4 \cdot 5 \cdot 6 \cdot 7 - \cancel{2 \cdot 3 \cdot 4 \cdot 5 \cdot 6}]$$

$$T_n = \frac{1}{5} [n \cdot (n+1) \cdot (n+2) \cdot (n+3) \cdot (n+4) - \cancel{(n-1) \cdot n \cdot (n+1) \cdot (n+2) \cdot (n+3)}]$$

$$S_n = \frac{1}{5} [n \cdot (n+1) \cdot (n+2) \cdot (n+3) \cdot (n+4) - 0]$$

Question



$1 \cdot 2 \cdot 5 + 2 \cdot 3 \cdot 6 + 3 \cdot 4 \cdot 7 + \dots$ Up to n terms

Question

$1 \cdot 3 + 3 \cdot 5 + 5 \cdot 7 + 7 \cdot 9 + \dots$ Up to n terms

$$T_n = (2n-1)(2n+1)$$

$$T_n = \frac{1}{6} \cdot (2n-1) \cdot (2n+1) \cdot 6$$

$$T_n = \frac{1}{6} \cdot (2n-1) \cdot (2n+1) \cdot [(2n+3) - (2n-3)]$$

$$T_n = \frac{1}{6} \left[\underbrace{(2n-1) \cdot (2n+1) \cdot (2n+3)}_{V_n} - \underbrace{(2n-3) \cdot (2n-1) \cdot (2n+1)}_{V_{n-1}} \right]$$

V_n V_{n-1}
 V_{n+1} V_n
 $n = n-1$
 $n = n+1$



$$T_n = (2n-3)(2n-1)(2n+1)(2n+3)$$

$n = n-1$ $n = n+1$

$$2 \cdot (n+1) + 1$$

$$2n + 2 + 1$$

$$\boxed{2n+3}$$

Question

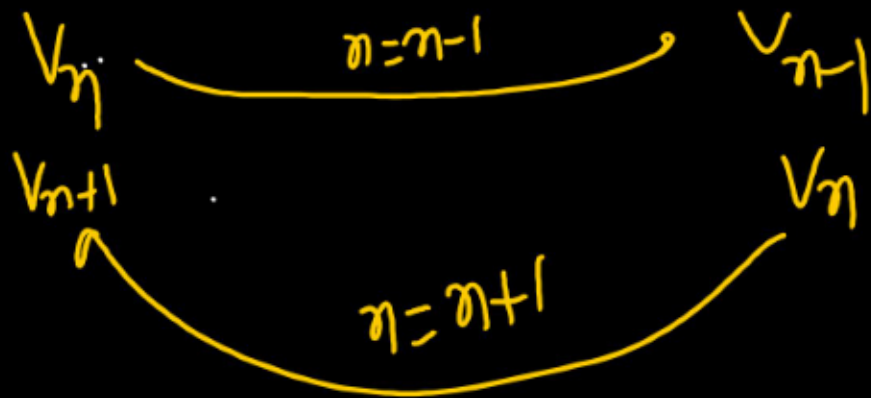
1.3 + 3.5 + 5.7 + 7.9 + Up to n terms

$$T_n = (2n-1)(2n+1)$$

$$T_n = \frac{1}{6} \cdot (2n-1) \cdot (2n+1) \cdot 6$$

$$T_n = \frac{1}{6} \cdot (2n-1) \cdot (2n+1) \cdot [(2n+3) - (2n-3)]$$

$$T_n = \frac{1}{6} \left[\underbrace{(2n-1) \cdot (2n+1) \cdot (2n+3)}_{V_n} - \underbrace{(2n-3) \cdot (2n-1) \cdot (2n+1)}_{V_{n-1}} \right]$$



$$T_1 = \frac{1}{6} [1 \cdot 3 \cdot 5 - (-1) \cdot 1 \cdot 3]$$

$$T_2 = \frac{1}{6} [3 \cdot 5 \cdot 7 - 1 \cdot 3 \cdot 5]$$

$$T_3 = \frac{1}{6} [5 \cdot 7 \cdot 9 - 3 \cdot 5 \cdot 7]$$

⋮

⋮

$$T_n = \frac{1}{6} [(2n-1)(2n+1)(2n+3) - (2n-3)(2n-1)(2n+1)]$$

$$S_n = \frac{1}{6} [(2n-1)(2n+1)(2n+3) - (-1) \cdot 1 \cdot 3]$$

Shortcut $\leftarrow \begin{array}{cc} | & | \\ 1 & n \end{array} \rightarrow$

$$S_n = \frac{1}{6} [(2n-1) \cdot (2n+1) \cdot (2n+3) - (-1) \cdot 1 \cdot 3]$$

Type : 2 (Splitting the n^{th} terms as a difference of two)

Here is a series in which each term is composed of the reciprocal of the product of r factors in A.P., the first factor of the several terms being in the same A.P.

$$S = \frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \frac{1}{3 \cdot 4} + \dots \text{ up to } n \text{ terms. or } S_n = \frac{1}{2} + \frac{1}{6} + \frac{1}{12} + \frac{1}{20} + \dots$$



$$T_n = \frac{1}{n \cdot (n+1)}$$

$$T_n = \frac{(n+1) - (n)}{n \cdot (n+1)}$$

$$T_n = \left[\frac{1}{\underbrace{n}_{V_n}} - \frac{1}{\underbrace{n+1}_{V_{n+1}}} \right]$$

$V_n \quad V_{n+1}$

$V_n \quad V_{n+1}$

$$T_1 = \left[\frac{1}{1} - \frac{1}{2} \right]$$

$$T_2 = \left[\frac{1}{2} - \frac{1}{3} \right]$$

$$T_3 = \left[\frac{1}{3} - \frac{1}{4} \right]$$

$$T_n = \left[\frac{1}{n} - \frac{1}{n+1} \right]$$

$$S_n = \left[1 - \frac{1}{n+1} \right]$$

$$S_n = \frac{n+1-1}{n+1}$$

$$S_n = \frac{n}{n+1}$$

shortcut

$$S_n = \left[1 - \frac{1}{n+1} \right]$$

$$S_\infty = 1$$

Question

$$\frac{1}{1 \cdot 2 \cdot 3 \cdot 4} + \frac{1}{2 \cdot 3 \cdot 4 \cdot 5} + \frac{1}{3 \cdot 4 \cdot 5 \cdot 6} + \dots \text{Up to } n \text{ terms}$$

$$T_n = \frac{1}{n \cdot (n+1) \cdot (n+2) \cdot (n+3)}$$

$$T_n = \frac{1}{3} \cdot \frac{3}{n \cdot (n+1) \cdot (n+2) \cdot (n+3)}$$

$$T_n = \frac{1}{3} \left[\frac{(n+3) - (n)}{n \cdot (n+1) \cdot (n+2) \cdot (n+3)} \right]$$

$$T_n = \frac{1}{3} \left[\underbrace{\frac{1}{n \cdot (n+1) \cdot (n+2)}}_{V_n} - \underbrace{\frac{1}{(n+1) \cdot (n+2) \cdot (n+3)}}_{V_{n+1}} \right]$$

$$T_1 = \frac{1}{3} \left[\frac{1}{1 \cdot 2 \cdot 3} - \frac{1}{2 \cdot 3 \cdot 4} \right]$$

$$T_2 = \frac{1}{3} \left[\frac{1}{2 \cdot 3 \cdot 4} - \frac{1}{3 \cdot 4 \cdot 5} \right]$$

$$T_3 = \frac{1}{3} \left[\frac{1}{3 \cdot 4 \cdot 5} - \frac{1}{4 \cdot 5 \cdot 6} \right]$$

$$\vdots$$
$$T_n = \frac{1}{3} \left[\frac{1}{n \cdot (n+1) \cdot (n+2)} - \frac{1}{(n+1) \cdot (n+2) \cdot (n+3)} \right]$$

$$S_n = \frac{1}{3} \left[\frac{1}{1 \cdot 2 \cdot 3} - \frac{1}{(n+1) \cdot (n+2) \cdot (n+3)} \right] \Rightarrow \boxed{S_{\infty} = \frac{1}{18}}$$

Question



$$\frac{1}{1 \cdot 3 \cdot 5} + \frac{1}{3 \cdot 5 \cdot 7} + \frac{1}{5 \cdot 7 \cdot 9} + \dots$$



In case a factor is missing in continued product in denominator we multiply and divide by that factor so that the in denominator we get a complete continued product.....

Question



$$\frac{1}{1 \cdot 2 \cdot 4} + \frac{1}{2 \cdot 3 \cdot 5} + \frac{1}{3 \cdot 4 \cdot 6} + \dots$$

Question



$\frac{1}{3^2-1} + \frac{1}{5^2-1} + \frac{1}{7^2-1} + \dots + \frac{1}{(201)^2-1}$ is equal to

- A. $\frac{101}{404}$
- ☒ B. $\frac{25}{101}$
- C. 4
- D. 6

$$T_n = \frac{1}{(2n+1)^2 - 1^2}$$

$$T_n = \frac{1}{4n^2 + 4n + 1 - 1}$$

$$T_n = \frac{1}{4 \cdot (n^2 + n)}$$

$$T_n = \frac{1}{4 \cdot n \cdot (n+1)}$$

$$T_n = \frac{1}{4} \left[\frac{(n+1) - (n)}{n \cdot (n+1)} \right]$$

$$T_n = \frac{1}{4} \left[\underbrace{\frac{1}{n}}_{V_n} - \underbrace{\frac{1}{n+1}}_{V_{n+1}} \right]$$

[JEE MAIN (March) 2021]

Shortcut

$$S_n = \frac{1}{4} \left[1 - \frac{1}{n+1} \right]$$

$$\begin{aligned} S_{100} &= \frac{1}{4} \left[1 - \frac{1}{100+1} \right] \\ &= \frac{1}{4} \left[1 - \frac{1}{101} \right] = \frac{1}{4} \times \frac{100}{101} \\ &= \frac{25}{101} \end{aligned}$$

Question



The sum of 10 terms of the series $\frac{3}{1^2 \times 2^2} + \frac{5}{2^2 \times 3^2} + \frac{7}{3^2 \times 4^2} + \dots$

[JEE MAIN (Aug) 2021]

A.

1

B.

$\frac{120}{121}$

C.

$\frac{99}{100}$

D.

$\frac{143}{144}$

$$T_n = \frac{(2n+1)}{n^2 \cdot (n+1)^2}$$

$$T_n = \frac{(n+1)^2 - n^2}{n^2 \cdot (n+1)^2}$$

$$T_n = \left[\underbrace{\frac{1}{n^2}}_{V_n} - \underbrace{\frac{1}{(n+1)^2}}_{V_{n+1}} \right]$$

$$n^2 \cdot (n^2 + 2n + 1)$$

$$S_n = \left[\frac{1}{1^2} - \frac{1}{(n+1)^2} \right]$$

$$S_{10} = \left[\frac{1}{1^2} - \frac{1}{(10+1)^2} \right] = \left[1 - \frac{1}{121} \right] = \frac{120}{121}$$

$$S_n = \frac{1}{1+1^2+1^4} + \frac{2}{1+2^2+2^4} + \frac{3}{1+3^2+3^4} + \frac{4}{1+4^2+4^4} + \dots \text{up to } n \text{ terms.}$$



Q) $\frac{1}{1-3 \cdot 1^2+1^4} + \frac{2}{1-3 \cdot 2^2+2^4} + \frac{3}{1-3 \cdot 3^2+3^4} + \dots \text{up to } n \text{ terms.}$

Q) $\frac{1}{1 \cdot (1+d)} + \frac{1}{(1+d)(1+2d)} + \frac{1}{(1+2d)(1+3d)} + \dots + \frac{1}{(1+9d)(1+10d)} = 5$, then $d = ?$

H.W.

$$S_n = \frac{1}{1+1^2+1^4} + \frac{2}{1+2^2+2^4} + \frac{3}{1+3^2+3^4} + \frac{4}{1+4^2+4^4} + \dots \text{ up to } n \text{ terms.}$$



$$\underline{\underline{2}} \quad T_n = \frac{n}{1+n^2+n^4}$$

$$T_n = \frac{n}{(n^2+1+n)(n^2+1-n)}$$

$$T_n = \frac{1}{2} \times \left[\frac{2n}{(n^2+1+n)(n^2+1-n)} \right]$$

$$T_n = \frac{1}{2} \left[\frac{(n^2+1+n) - (n^2+1-n)}{(n^2+1+n)(n^2+1-n)} \right]$$

$$n^4+n^2+1 = (n^2+1+n)(n^2+1-n)$$

$$n^4+n^2+1+n^2-n^2$$

$$\underbrace{n^4+2n^2+1}_{(n^2+1)^2} - n^2$$

$$(n^2+1+n)(n^2+1-n)$$

$$\therefore S_n = \frac{1}{2} \left[\frac{1}{1} - \frac{1}{1+n^2+n} \right]$$

$$T_n = \frac{1}{2} \left[\frac{1}{\underbrace{n^2+1-n}_{\uparrow}} - \frac{1}{n^2+1+n} \right]$$

$$T_n = \frac{1}{2} \left[\frac{1}{1+n^2-n} - \frac{1}{1+n^2+n} \right]$$

$$T_n = \frac{1}{2} \left[\frac{1}{\underbrace{(1-n) \cdot (n+1)}_{V_n}} - \frac{1}{\underbrace{(1+n) \cdot (n+1)}_{V_{n+1}}} \right]$$

$$S_n = \frac{1}{1 \cdot 3} + \frac{1}{2 \cdot 4} + \frac{1}{3 \cdot 5} + \frac{1}{4 \cdot 6} + \dots$$

$$T_n = \frac{1}{n \cdot (n+2)}$$

$$T_n = \frac{1}{2} \times \frac{2}{n \cdot (n+2)}$$

$$T_n = \frac{1}{2} \left[\frac{(n+2) - n}{n \cdot (n+2)} \right]$$

$$T_n = \frac{1}{2} \left[\underbrace{\left(\frac{1}{n} \right)}_{\downarrow n} - \underbrace{\frac{1}{n+2}}_{\downarrow n+2} \right]$$

$$T_1 = \frac{1}{2} \left[\frac{1}{1} - \cancel{\frac{1}{3}} \right]$$

$$T_2 = \frac{1}{2} \left[\frac{1}{2} - \cancel{\frac{1}{4}} \right]$$

$$T_3 = \frac{1}{2} \left[\cancel{\frac{1}{3}} - \frac{1}{5} \right]$$

$$T_4 = \frac{1}{2} \left[\cancel{\frac{1}{4}} - \frac{1}{6} \right]$$

$$\vdots$$

$$T_{n-1} = \frac{1}{2} \left[\cancel{\frac{1}{n-1}} - \frac{1}{n+1} \right]$$

$$T_n = \frac{1}{2} \left[\cancel{\frac{1}{n}} - \frac{1}{n+2} \right]$$

$$S_n = \frac{1}{2} \left[1 + \frac{1}{2} - \left(\frac{1}{n+1} + \frac{1}{n+2} \right) \right]$$

Shortcut



$$S_n = \frac{1}{2} \left[\left(1 + \frac{1}{2} \right) - \left(\frac{1}{n+2} + \frac{1}{n+1} \right) \right]$$





Topic: Homework-1



- DPP

DPP NO	Question No
DPP -1	
DPP -2	

- Module
- Exercise Name

Module Page No	Question No
Page No- 21	
Page No -22	



Next Lecture's Goal

Chap Name

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1. Sub Top

..

2. Sub Top

..

3. Sub Top

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4. Sub Top

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5. Sub Top

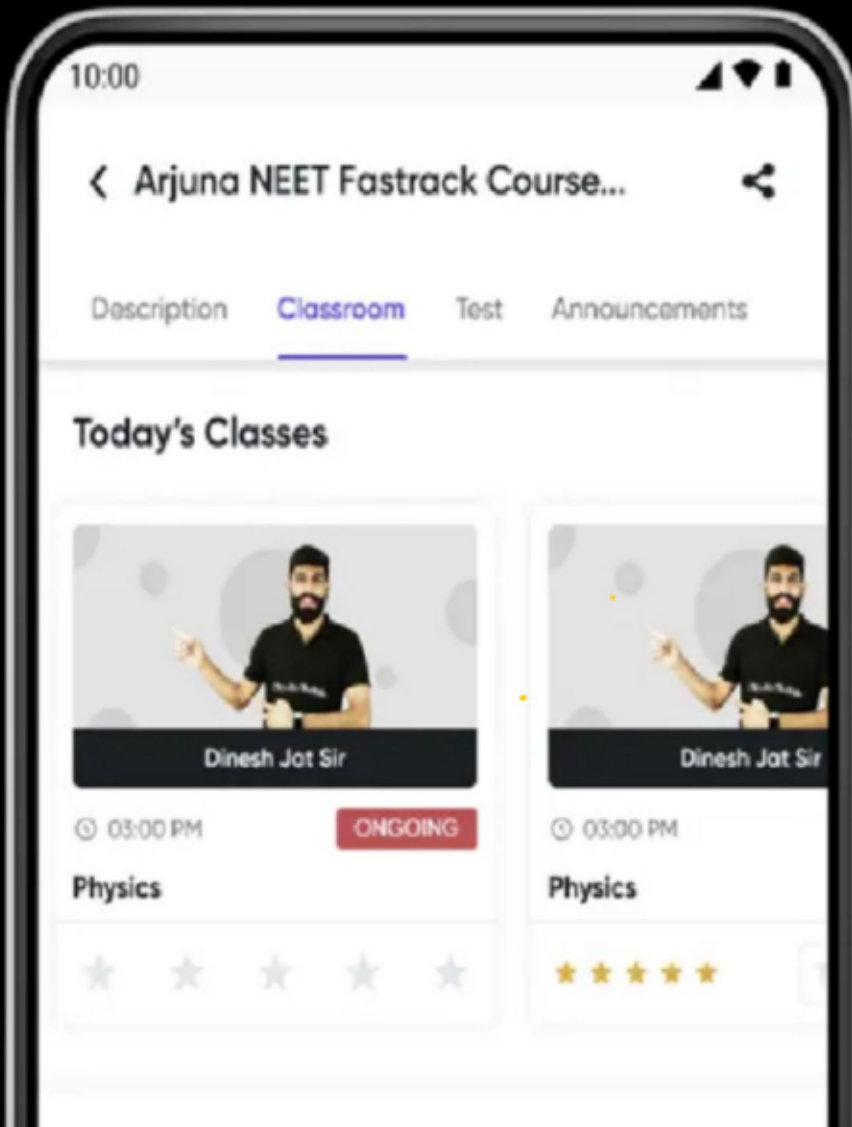
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Solve the DPP and check Solution



"SCAN" to join our "TELEGRAM" channel

VIDYAPEETH

WORK, POWER AND ENERGY

DPP-1 (JAP/046)

[Introduction, Definition of work, work done by constant force, Area under force-displacement curve]

1. A particle moves from position $\vec{r}_1 = 3\hat{i} + 2\hat{j} - 6\hat{k}$ to position $\vec{r}_2 = 14\hat{i} + 13\hat{j} + 9\hat{k}$ under the action of force $4\hat{i} + \hat{j} + 3\hat{k}$ N. The work done by this force will be

(A) 100 J
(B) 50 J

(A) 8×10^{-2} joules
(B) 16×10^{-2} joules
(C) 4×10^{-2} joules

Thank You!!!!

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